
x
[B,n,H]
Wqkv
Q, K, V
[B,n,H]
K̂, L2
pool keys
k_pool (4:1)
cluster
Ward buckets
c_b (Alg. 1)
max_h q̂·c_b
top-p buckets →
k long-range
keys
expand
K,V
QKᵀ
J = sinks ∪ k
∪ window
QKᵀ
sparse scores
= QKᵀ over J
softmax
softmax
Wo
O =
Wo·(softmax·V)
[B,n,H]